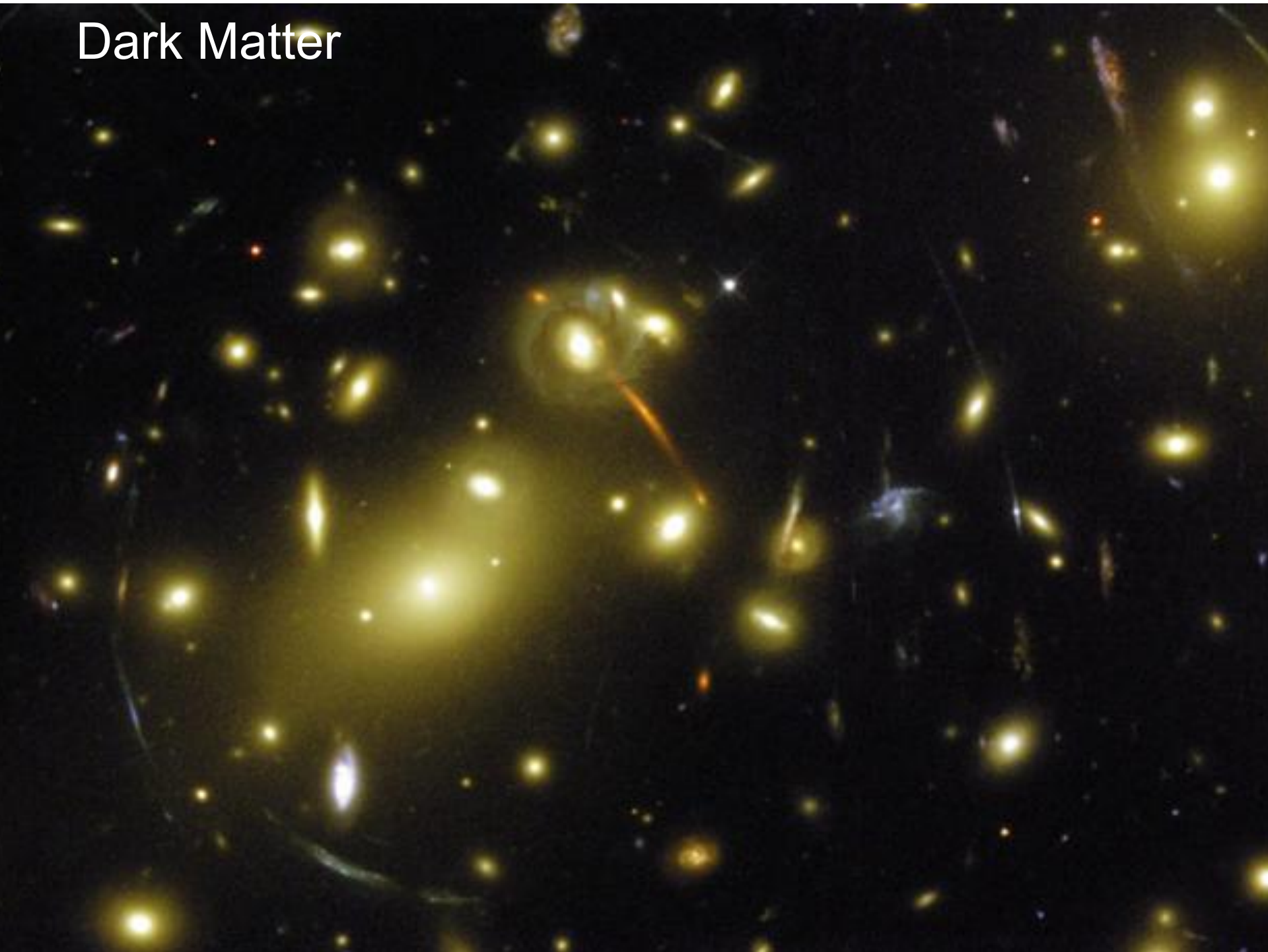
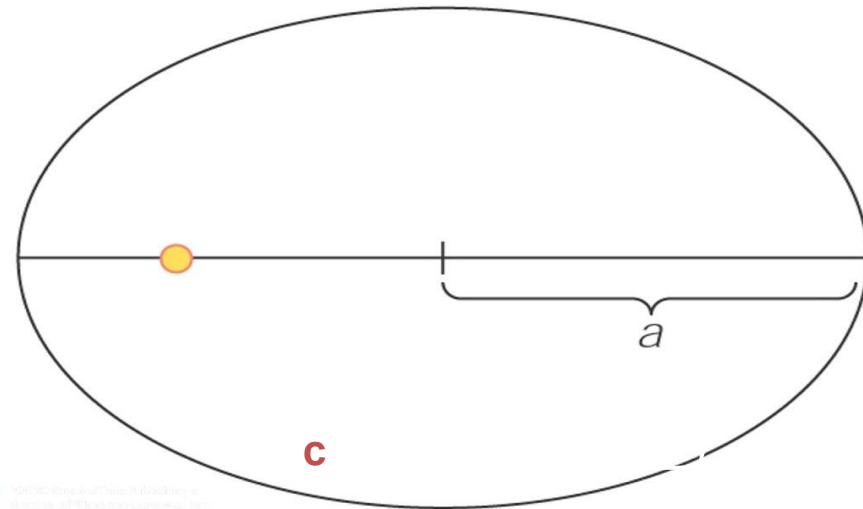


Dark Matter

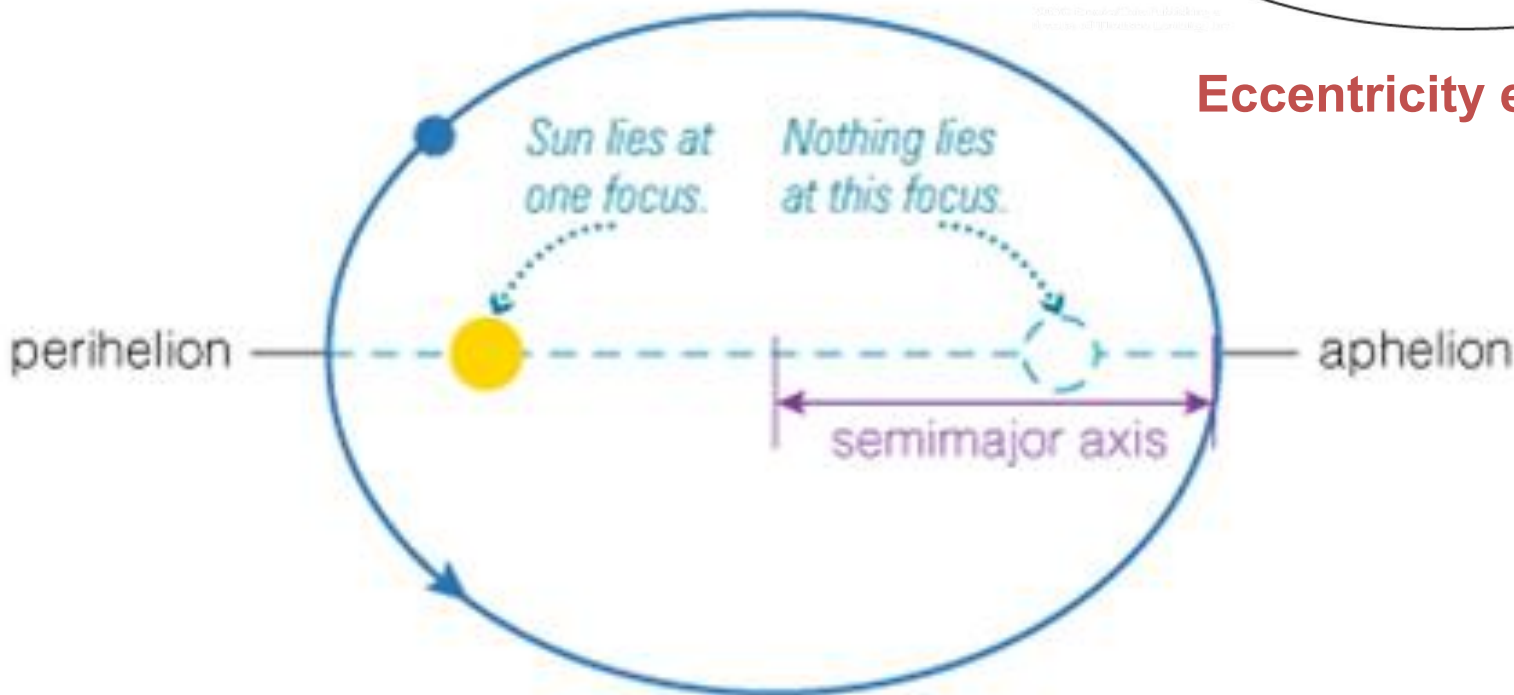


Kepler's 1st Law

Law #1: The orbits of the planets are ellipses with the Sun at one focus.



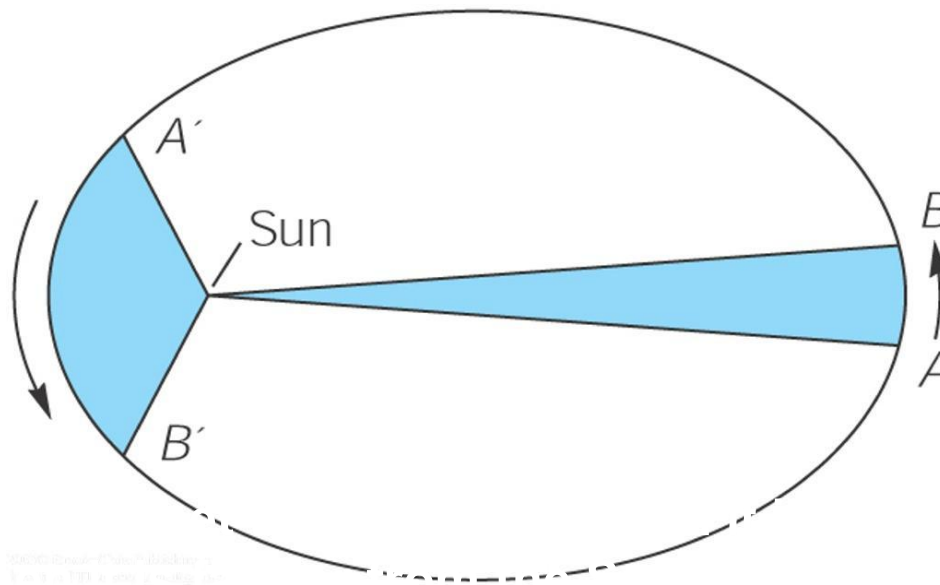
$$\text{Eccentricity } e = c/a$$



Kepler's 2nd Law

Law #2: A line from a planet to the sun sweeps over equal areas in equal intervals of time.

(When it's closer to the Sun, a planet moves faster)



Kepler's 3rd Law

A planet's orbital period (P) squared is proportional to its average distance from the sun (a) cubed:

Closer orbits go faster

$$P_y^2 = a_{AU}^3$$

(P_y = period in years; a_{AU} = distance in AU)

<http://www.solarsystemscope.com/>

<http://astro.unl.edu/naap/pos/animations/kepler.html>

Concept Quiz

Which planet will orbit in the smallest amount of time?

- a) Something with the mass of Jupiter at the orbital distance of Jupiter
- b) Something with the mass of Earth at the orbital distance of Jupiter
- c) Something with the mass of Earth at the orbital distance of Earth
- d) Something with the mass of Jupiter at the orbital distance of Earth
- e) c) and d) would be equal

Concept Quiz

Which planet will orbit in the smallest amount of time?

- a) Something with the mass of Jupiter at the orbital distance of Jupiter
- b) Something with the mass of Earth at the orbital distance of Jupiter
- c) Something with the mass of Earth at the orbital distance of Earth
- d) Something with the mass of Jupiter at the orbital distance of Earth
- e) c) and d) would be equal

Concept Quiz

How long does it take Mars to orbit the Sun?

- a) More than a year
- b) Less than a year
- c) One year

Concept Quiz

How long does it take Mars to orbit the Sun?

- a) More than a year
- b) Less than a year
- c) One year

Orbits, mass, and radius

Orbital speed is related to the mass interior to the orbit, and the orbital radius

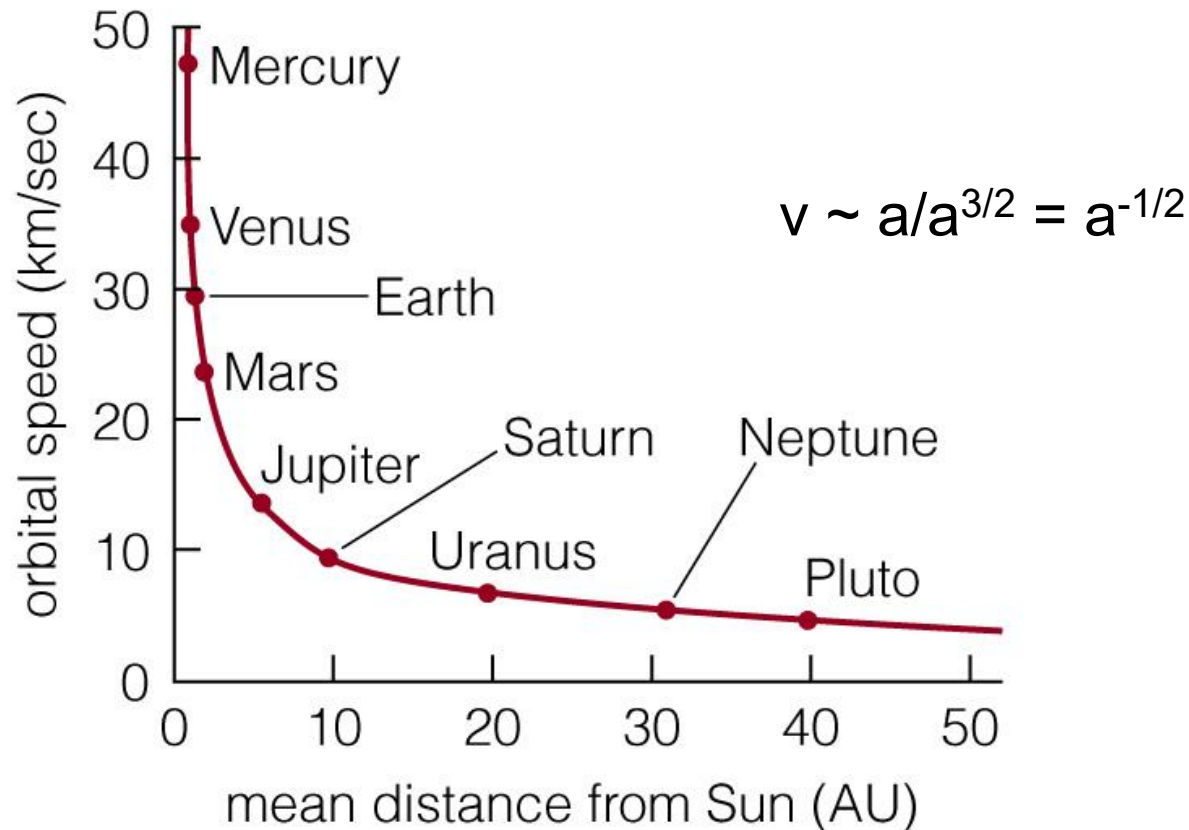
<https://phet.colorado.edu/en/simulations/gravity-and-orbits>

$$\text{If } P^2 = a^3, P = a^{3/2}$$

$$v = 2\pi a/P$$

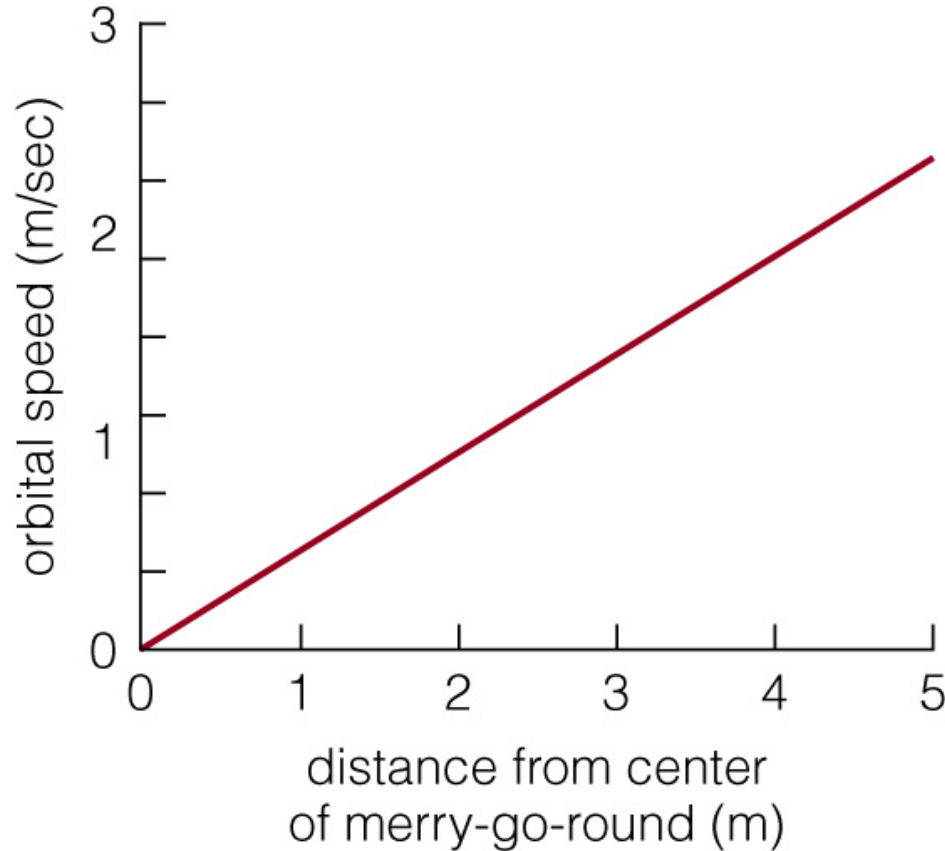
$$\rightarrow v \sim a/a^{3/2} = a^{-1/2}$$

Solar System Rotation Curve



- Solar system's rotation curve declines because Sun has almost all the mass
- Called "Keplarian" since this is what is expected from Kepler's 3rd law

Merry-go-round Rotation Curve



All periods are the same

$$v = 2\pi a/P$$

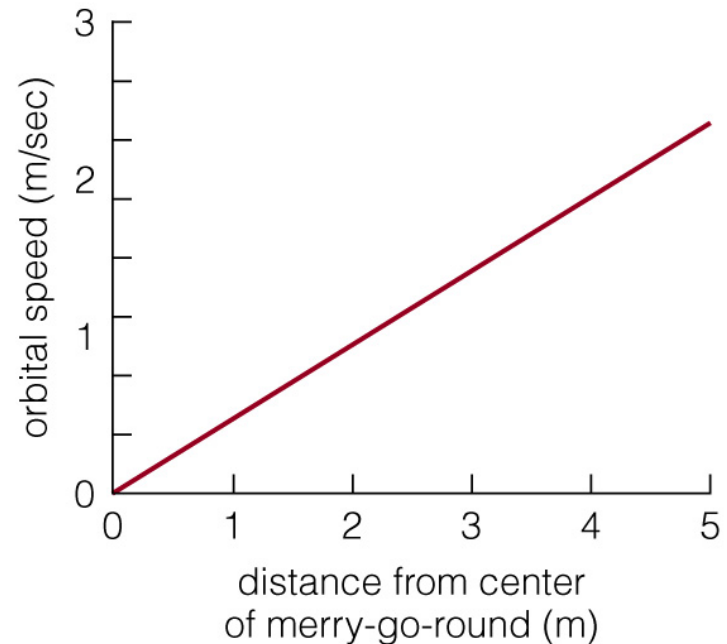
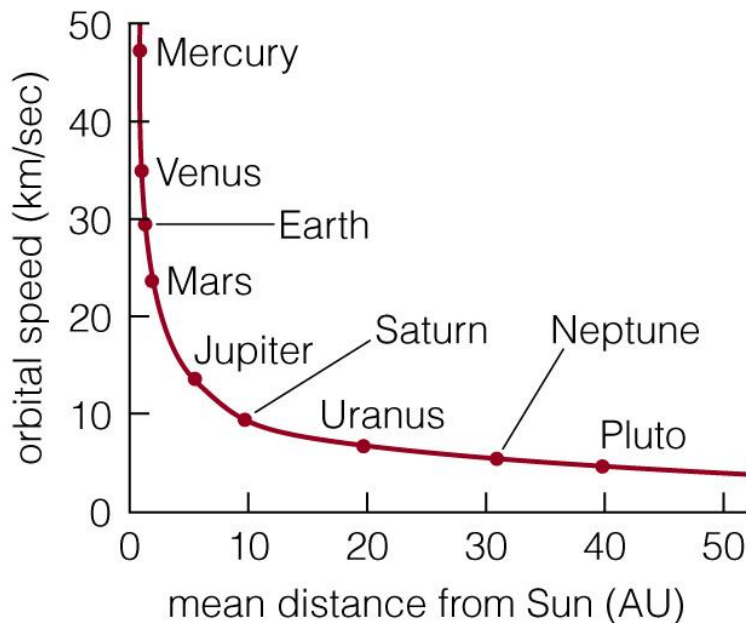
$$\rightarrow v \sim a$$

Rotation curve of merry-go-round rises with radius because mass is spread out with radius. Called “Solid body”

Concept Quiz

What will the rotation curve of our Galaxy look like?

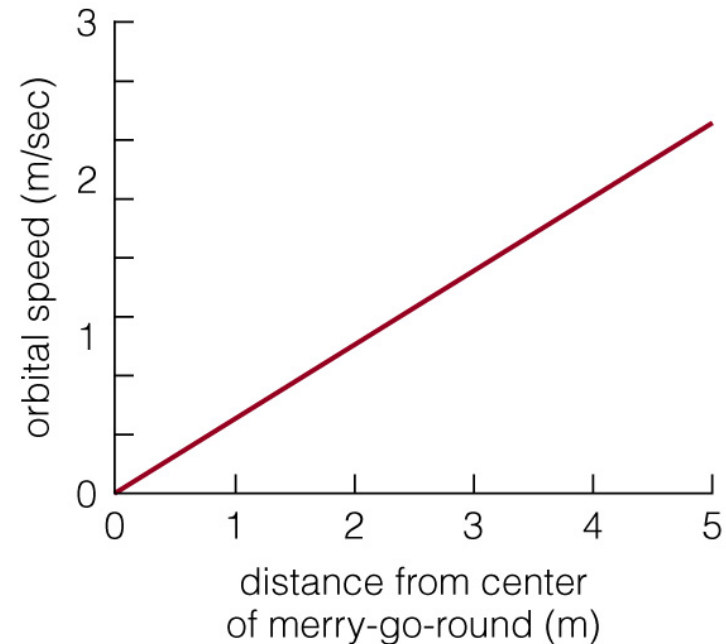
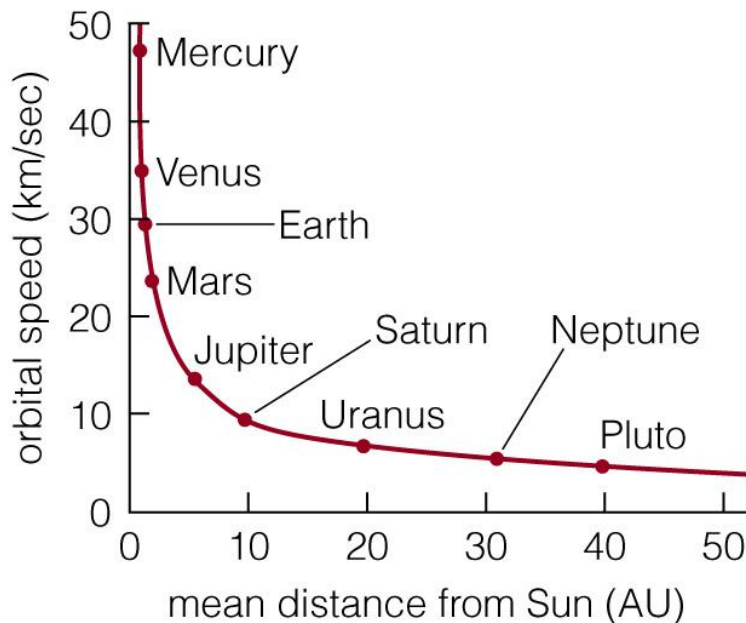
- A) Like that of the Solar System
- B) Like that of the Merry-go-round



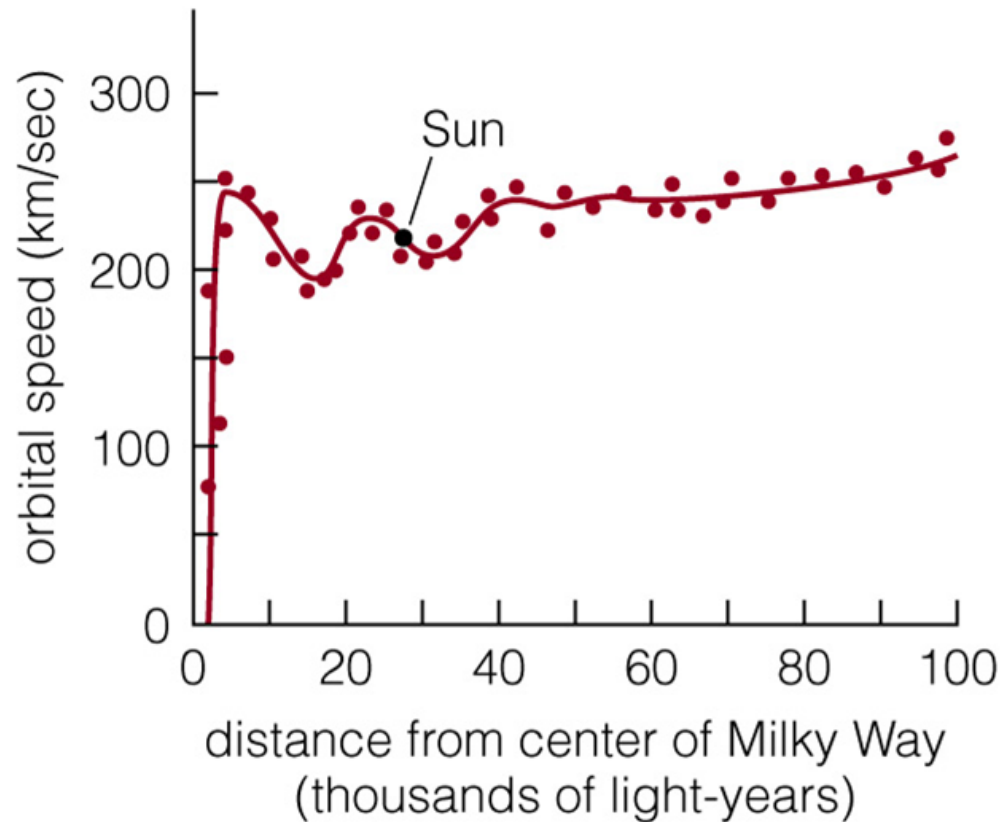
Concept Quiz

What will the rotation curve of our Galaxy look like?

- A) Like that of the Solar System
- B) Like that of the Merry-go-round
- C) Neither! Ha!

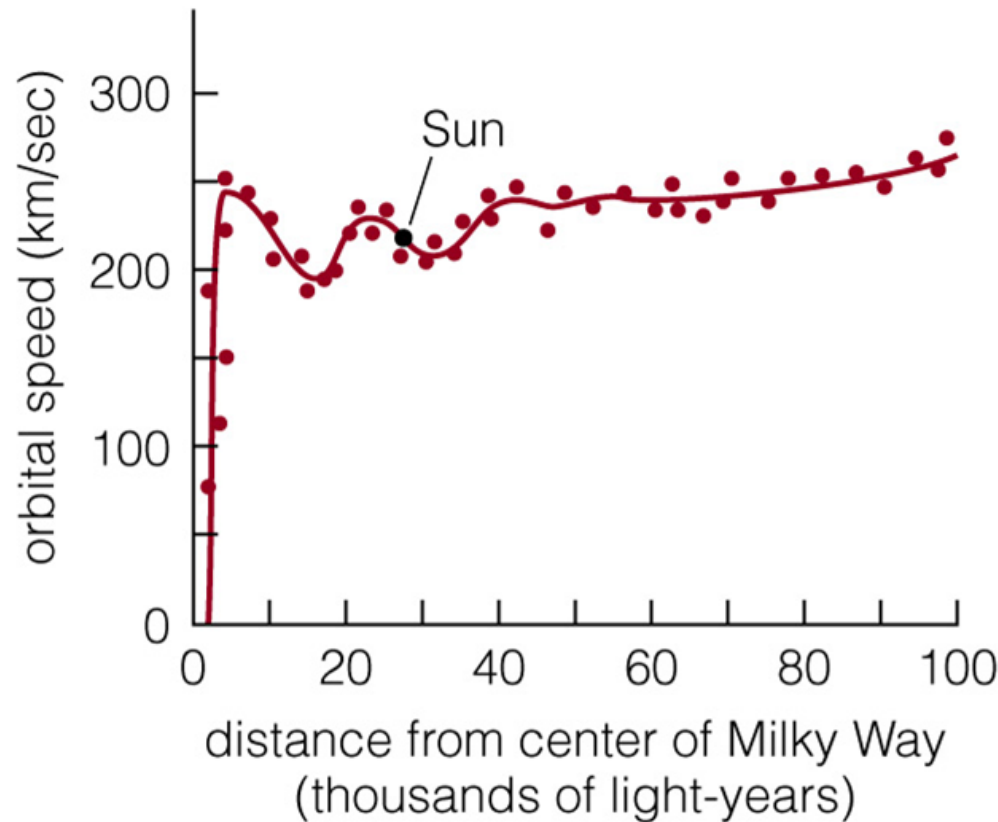


Milky Way Rotation Curve



Mass must be more spread out than in solar system,
but not as much as for a merry-go-round

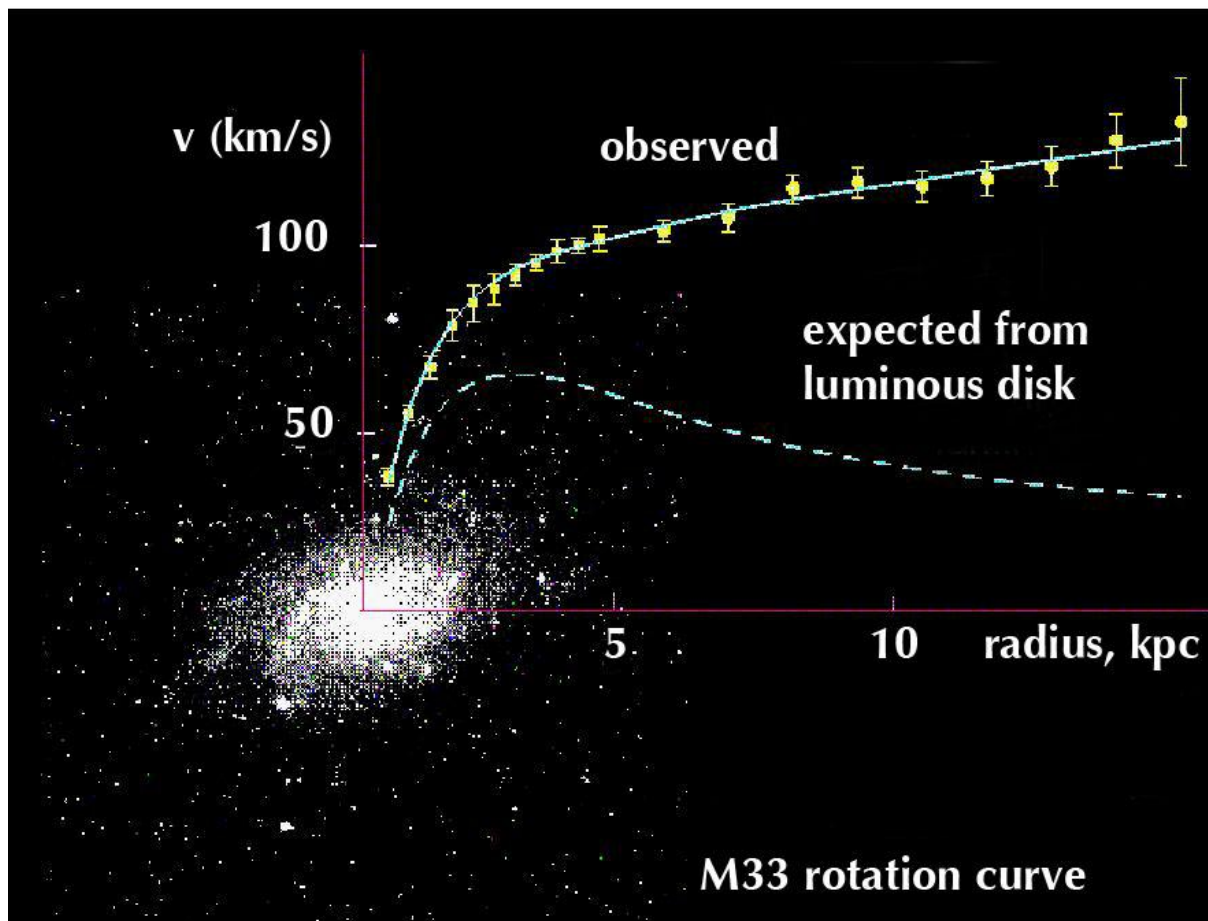
Milky Way Rotation Curve



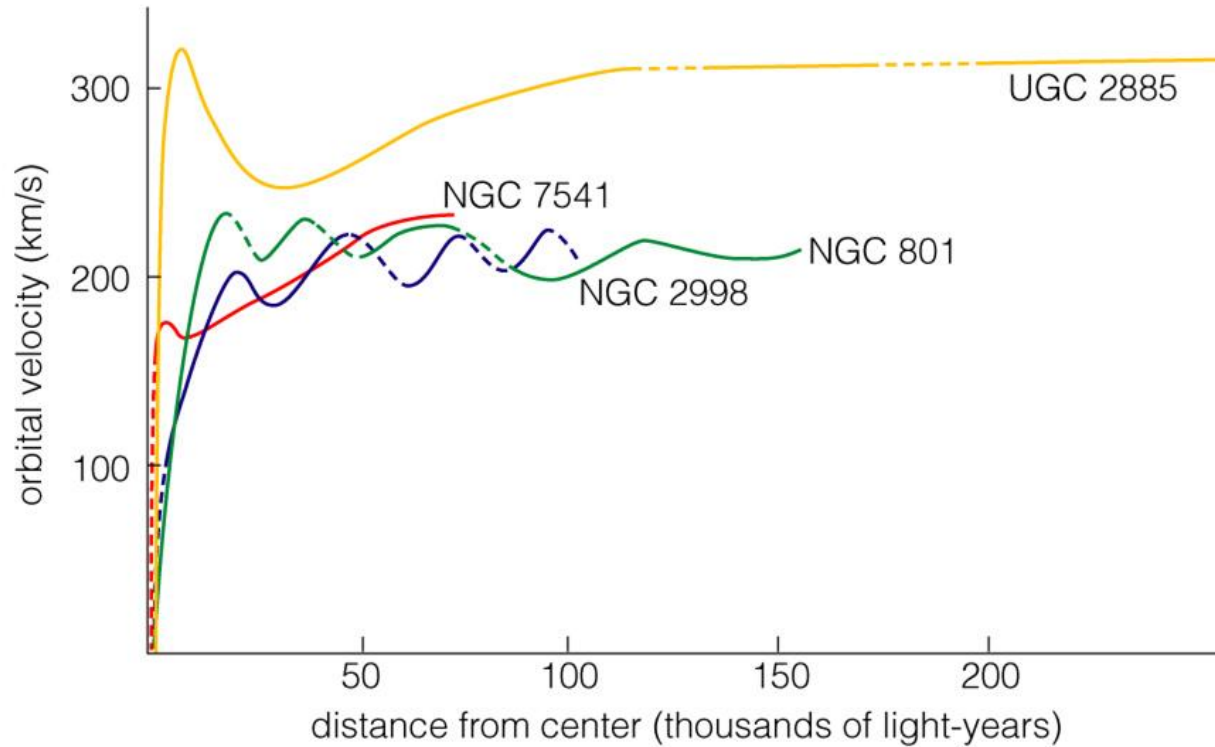
But from visible matter, we would expect something more like what we see in the Solar System because most of the mass is in the center of the Galaxy.

Evidence for Dark Matter

We hypothesize that dark matter exists from comparisons between the mass of the matter we can see and the mass calculated from gravity



Dark Matter in Spirals

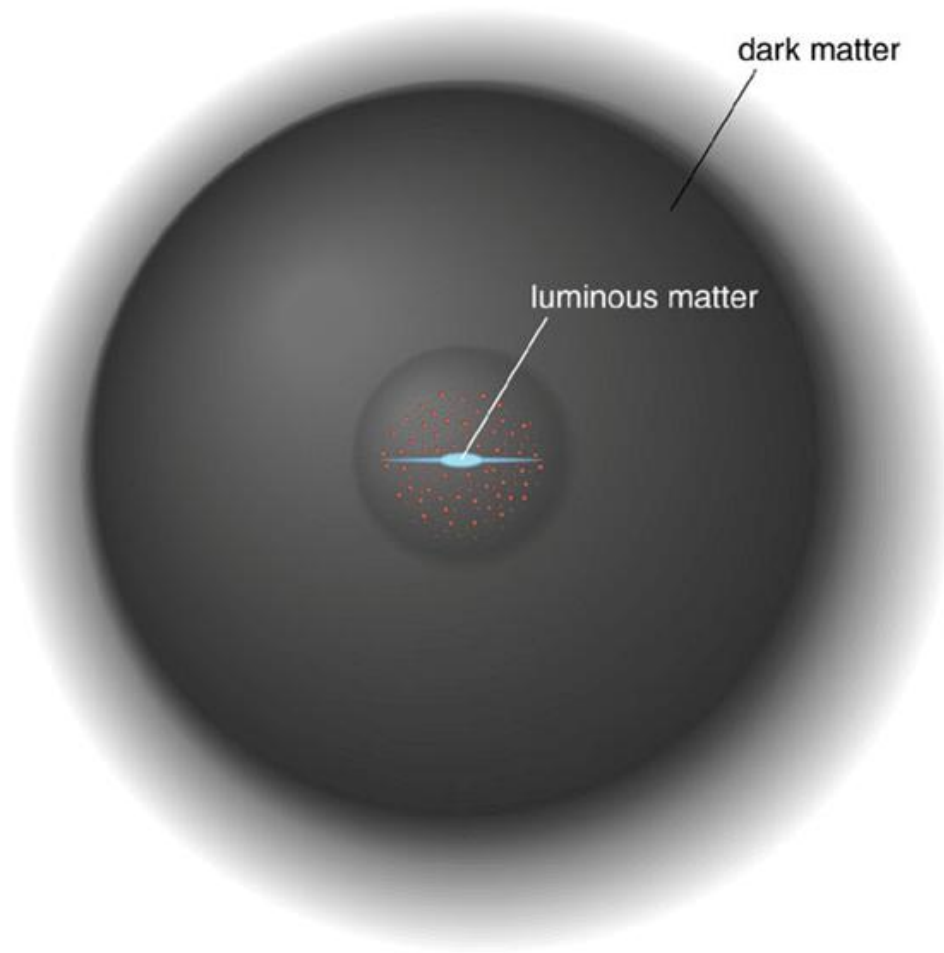


- Spiral galaxies all tend to have flat rotation curves
-> Lots of Dark Matter!

More Questions

- It's not obvious from the plot, but the Milky Way rotation curve rises linearly within the bulge. What can we say about the distribution of mass here – what example is it like?
- If the dark matter ended half way out from the center, what would the Milky Way rotation curve look like?

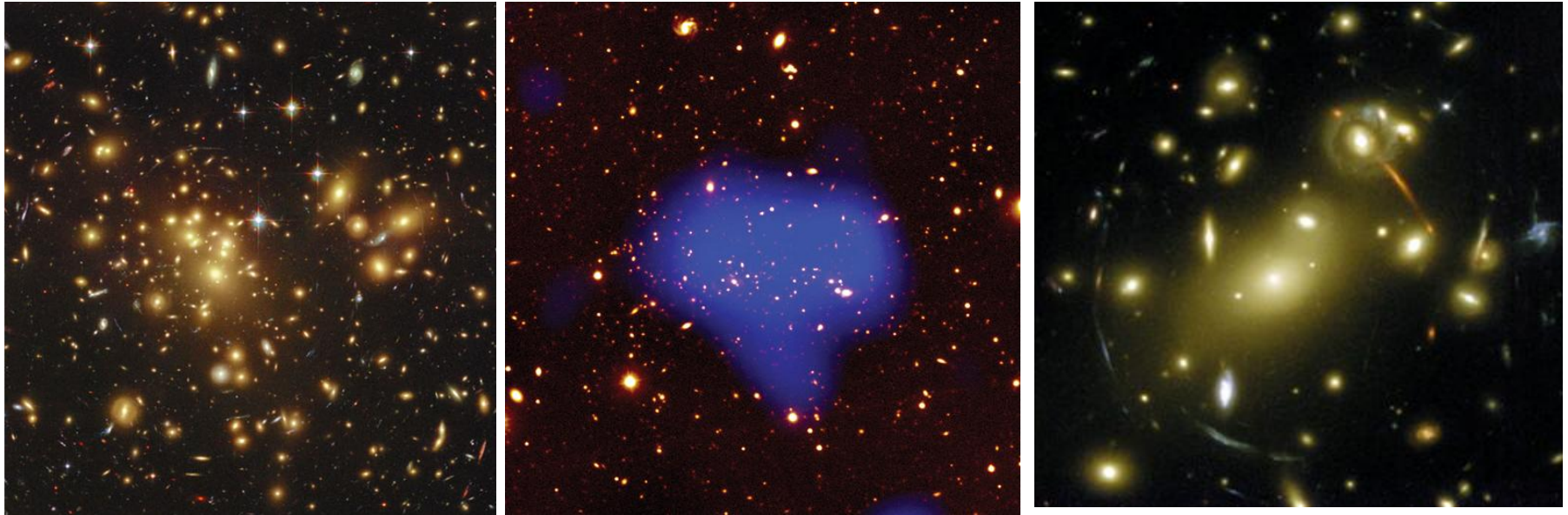
Dark Matter



The visible portion of a galaxy lies deep in the heart of a large halo of dark matter

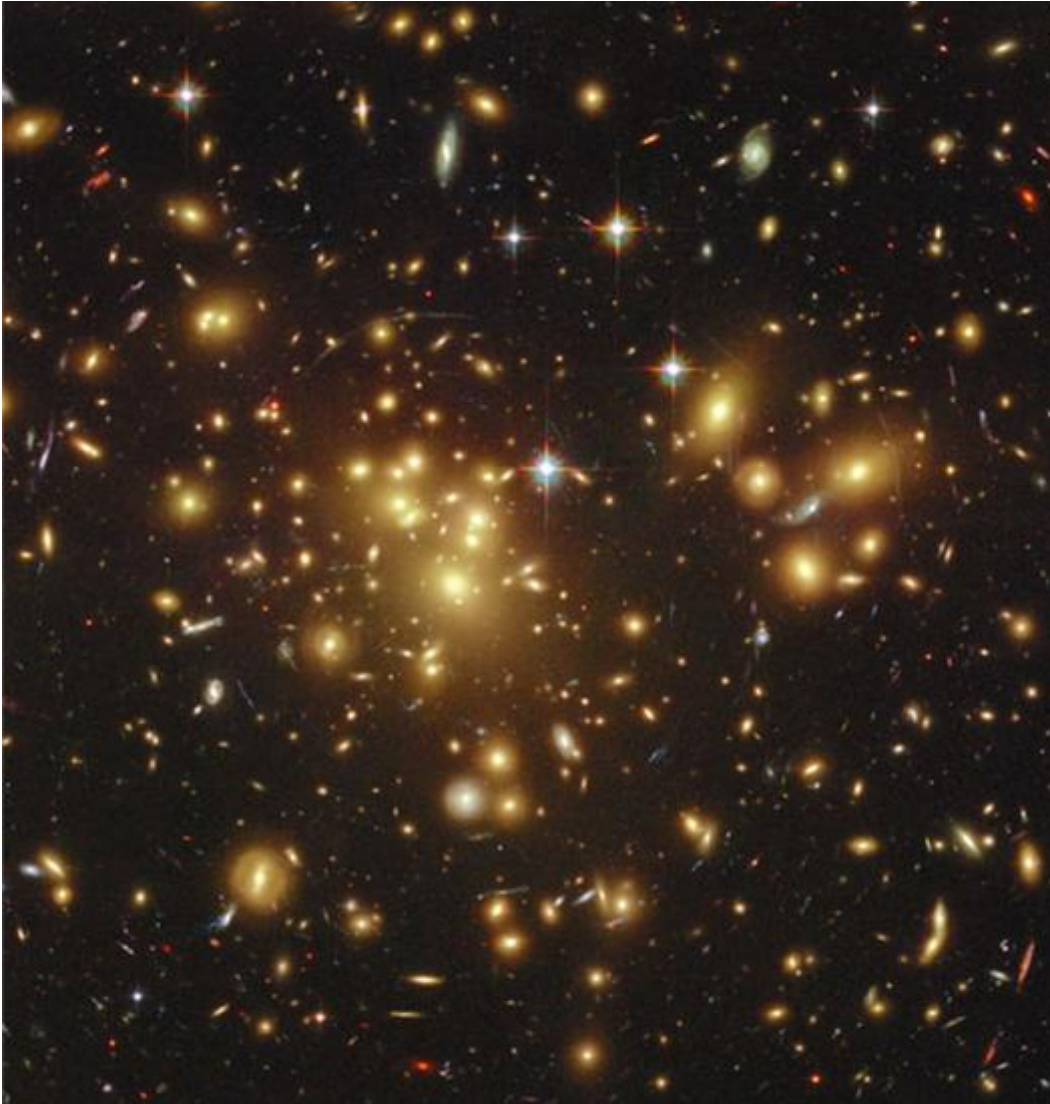
Dark Matter
provides the missing mass

Dark Matter in Clusters



Clusters are large groups of galaxies
There are three ways to find cluster mass

1) Velocities of galaxies

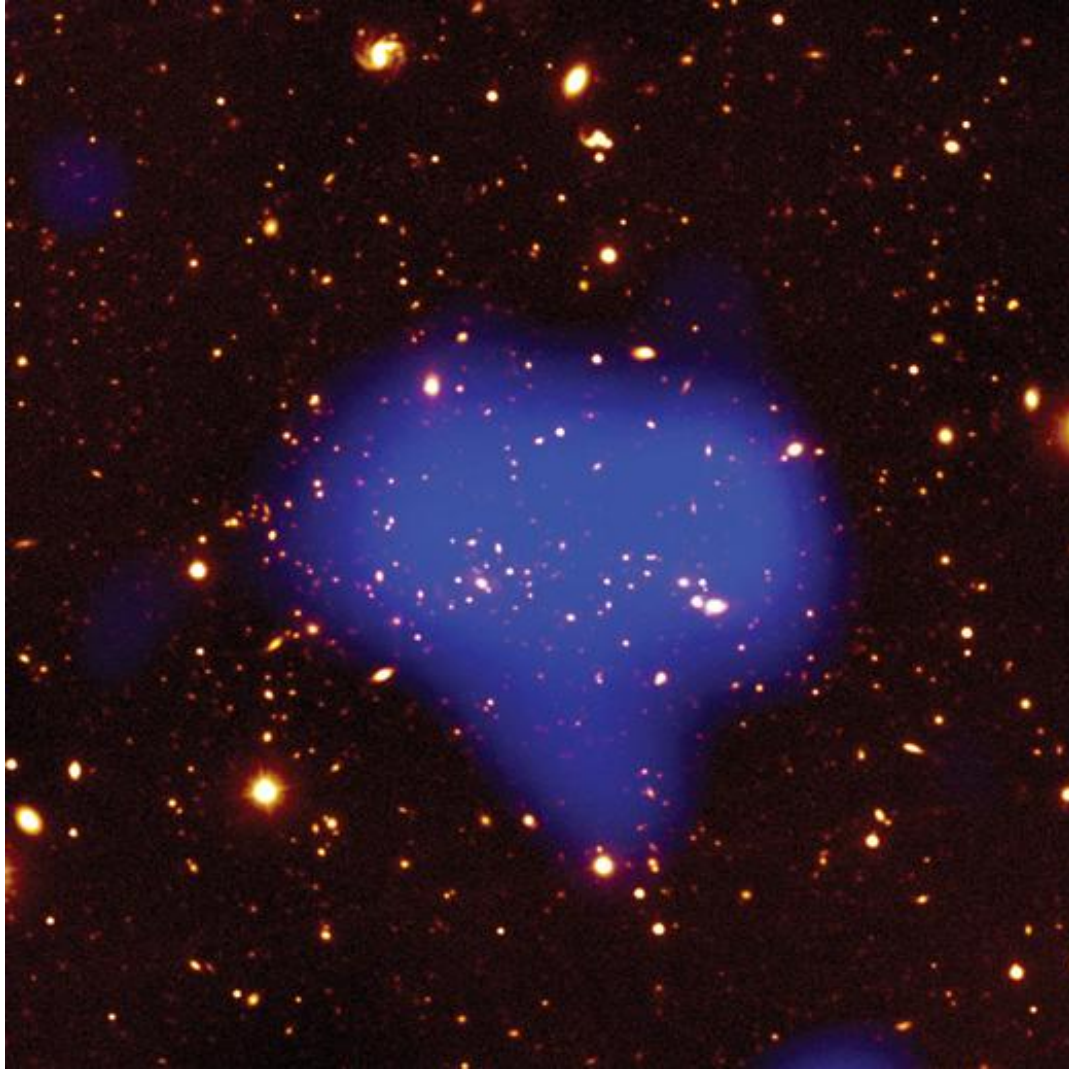


The galaxies are moving too fast, indicating unseen mass

The mass we calculate from galaxy motions in a cluster is about

50 times larger than the mass in stars!

2) Mass traced through X-rays

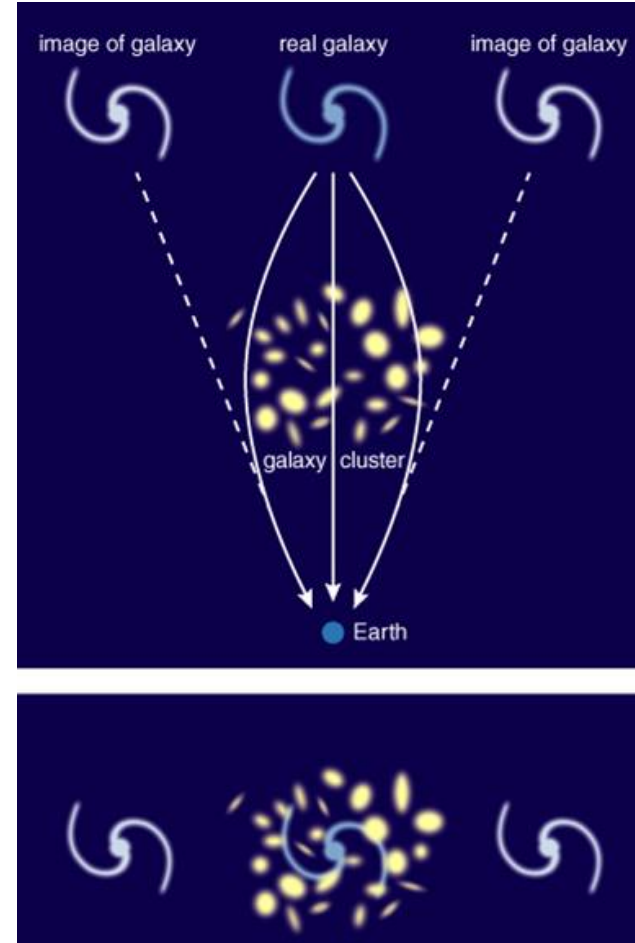
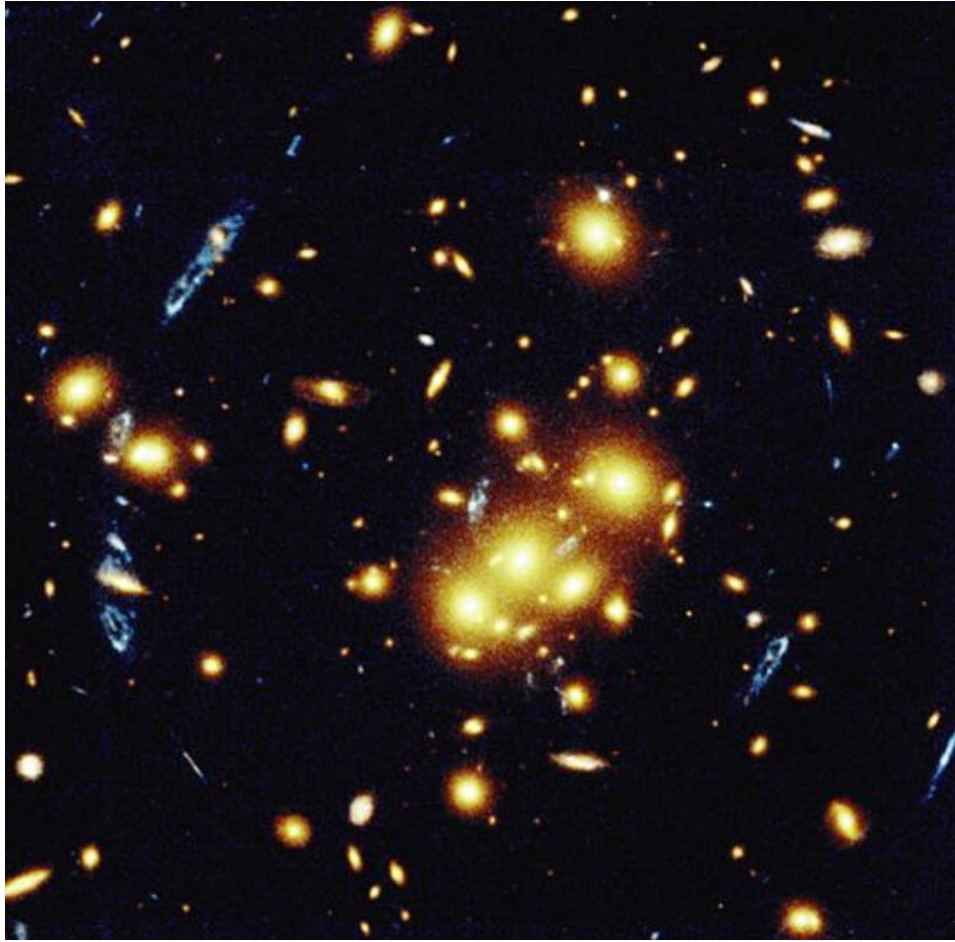


Clusters contain large amounts of X-ray emitting hot gas

Temperature of hot gas (particle motions) tells us cluster mass. The gas is hotter than it would be if there was no dark matter.

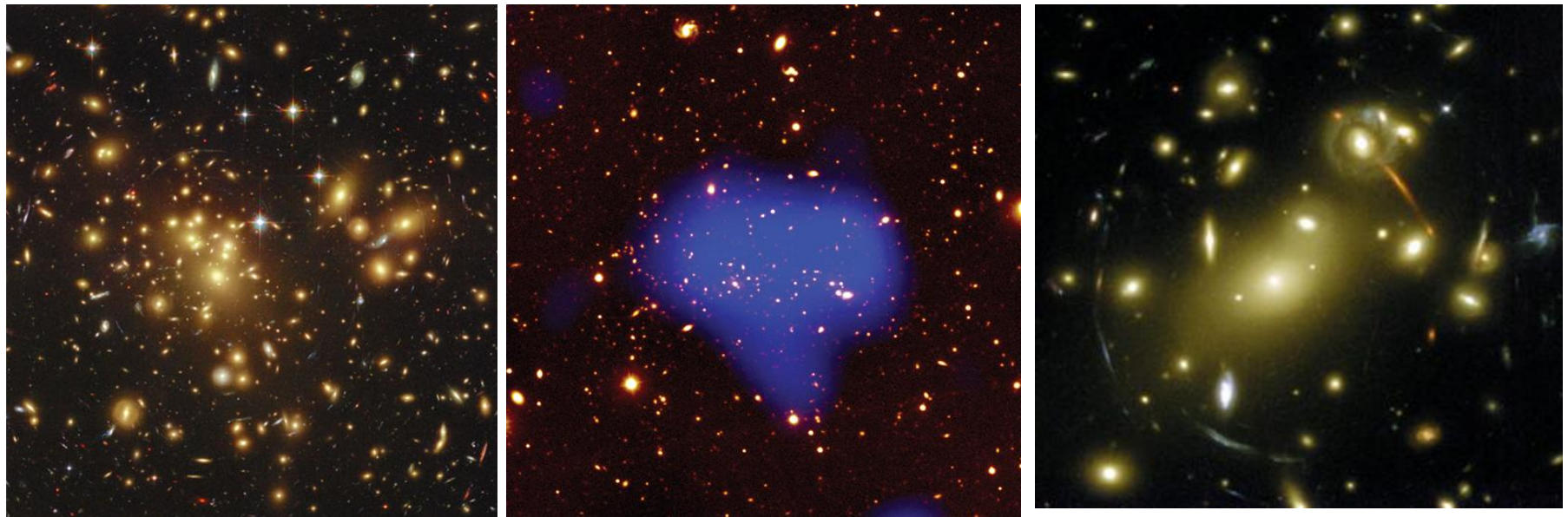
85% dark matter
13% hot gas
2% stars

3) Gravitational Lensing



Gravitational lensing, the bending of light rays by gravity, can also tell us a cluster's mass
 Light is bending more than expected

Cluster Masses



All three methods of measuring cluster mass indicate similar amounts of dark matter

But Does Dark Matter Really Exist???

Something is needed to explain the discrepancy between the mass measured through gravity and the mass directly observed.

- 1) Dark matter really exists, and we are observing the effects of its gravitational attraction
- 2) Something is wrong with our understanding of gravity, causing us to mistakenly infer the existence of dark matter

Because gravity is so well tested, most (but not all!) astronomers prefer option #1

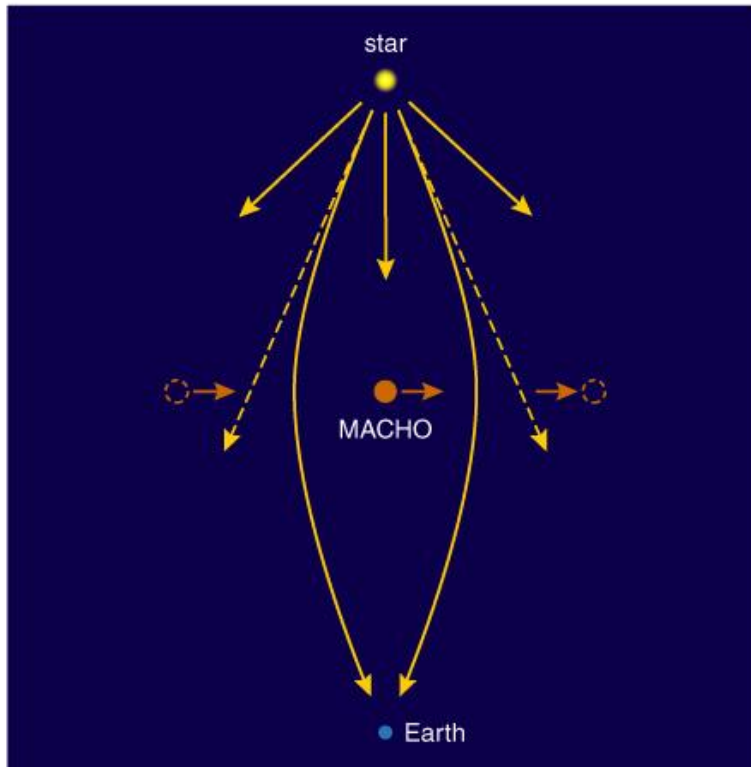
How dark is dark matter?

... not as bright as a star.

Two Basic Options

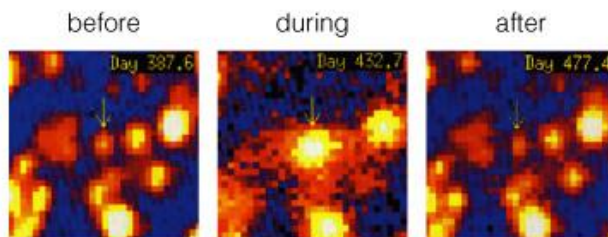
- Ordinary Dark Matter (MACHOS)
 - Massive Compact Halo Objects:
dead or failed stars in halos of galaxies – basically brown dwarfs
- Extraordinary Dark Matter (WIMPS)
 - Weakly Interacting Massive Particles:
a new type of particle – possibly similar to a neutrino

Evidence for MACHOS



MACHOs occasionally make other stars appear brighter through lensing

... but there are not enough lensing events to explain all the dark matter



Evidence for WIMPs

- There's not enough ordinary matter
- WIMPs could be left over from Big Bang
- Models involving WIMPs explain how galaxy formation works